Terminal Object 1

$\forall A.\ \exists$ a unique mapping $A \to 1$

1) $\langle\rangle : A \to 1$

2) $$\frac{M : A \to 1}{M = \langle\rangle : A \to 1}$$

$$\frac{}{\Gamma \vdash \langle\rangle : T}\ (T\text{-}I)$$

$$\frac{\Gamma \vdash M : T}{\Gamma \vdash M \equiv \langle\rangle : T}\ (\eta T)$$

Product $A \times B$ "final span between A and B"

1) existence

$\forall A B.\ \exists C = A \times B$ s.t.

$fst : A \times B \to A$

$snd : A \times B \to B$

$$\frac{\Gamma \vdash M : A \wedge B}{\Gamma \vdash fst(M) : A}\ \wedge E_1$$

$$\frac{\Gamma \vdash M : A \times B}{\Gamma \vdash snd(M) : B}\ \wedge E_2$$

$$\frac{M : D \to A \quad N : D \to B}{\langle M, N\rangle : D \to A \times B}$$

$fst \circ \langle M, N\rangle = M : D \to A\ (\beta X_1)$

$snd \circ \langle M, N\rangle = N : D \to B\ (\beta X_2)$

2) uniqueness

$$\frac{P : D \to A \times B \quad fst \circ P = M \quad snd \circ P = N}{P = \langle M, N\rangle : D \to A \times B}\ (\eta X)$$

$$\langle fst, snd \rangle = id$$

$\forall MNP.$ $\langle M, N \rangle \circ P = \langle M \circ P, N \circ P \rangle$

$$\begin{tikzcd}
D \arrow[d, "P"'] \arrow[dd, bend right=60, "M"'] \arrow[dd, bend left=60, "N"] \\
A \times B \arrow[d, "fst"] \arrow[d, shift left=6, "snd"] \\
A \quad B
\end{tikzcd}$$

$$\Rightarrow \langle fst \circ P, snd \circ P \rangle = \langle fst, snd \rangle \circ P = id \circ P = P$$

<u>Exponential</u> B^A

1) existence

$\forall A B, \exists B^A$ s.t. $B^A \times A \xrightarrow{ap} B$

$$\begin{tikzcd}
C \arrow[d, "\lambda(h)"'] \arrow[d, shift left=6, "g"] \\
B^A
\end{tikzcd}
\qquad
\begin{tikzcd}
C \times A \arrow[d, "\langle \lambda(h) \circ fst, snd \rangle"'] \arrow[d, shift left=6, "g \times id_A"] \arrow[dr, "h"] & \\
B^A \times A \arrow[r, "ap"] & B
\end{tikzcd}$$

$= \lambda(h) \times id_A$

$$ap \circ (\lambda(h) \times id_A) = ap \circ \langle \lambda(h) \circ fst, snd \rangle = h$$

2) uniqueness

$$\frac{ap \circ (g \times id_A) = h : C \times A \to B}{g = \lambda(h) : C \to B^A} \; (\eta)$$

$$\frac{\Gamma, x:A \vdash M:B}{\Gamma \vdash \lambda x.M : B^A} \supset I$$

$\Gamma \vdash M : A$ | $M : \overline{\Gamma} \to A$

$$\left(\left(\left(\emptyset \times A_n\right) \times A_{n-1}\right) \times \cdots\right) \times A_0$$

$$A_i = \mathrm{Snd} \times \mathrm{fst}^{(i)}(\overline{\Gamma})$$

"De Bruijn Index"

$$\mathrm{Hom}_C(Z \times B, C) \simeq \mathrm{Hom}_C(Z, C^B)$$

$$\begin{tikzcd}
C^B & C^B \times B \arrow[r, "\varepsilon"] & C \\
Z \arrow[u, "\tilde{f}"] & Z \times B \arrow[u, "\tilde{f} \times 1_B"] \arrow[ur, "f"'] &
\end{tikzcd}$$